Nuts and Bolts of New Ventures

Joseph Hadzima

MIT 15.393 OpenCourseWare entrepreneurship course
Lecture notes organized by LazyingArt LLC with Video2Book

Original lectures by Joseph Hadzima for MIT OpenCourseWare. Lecture notes organized by LazyingArt LLC.

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Chapter 1

Introduction: Most Startups Fail; How to Improve Your Odds

Overview. Joseph Hadzima begins this MIT OpenCourseWare course the right way: not with a doctrine, not with a slogan, but with a room full of motives and questions. He first asks why we are here, then what we do not know, then why we would attempt something so painful given such poor odds. Only after that does he permit himself equations, frameworks, and diagrams. These notes follow that sequence closely. They are curated by LazyingArt LLC from the source lecture, and they are meant to read as companion notes for a practical course in venture formation rather than as startup mythology.

1.1 Why We Are Here

The lecture opens with a guide's promise. We are beginning a six-session journey through the nuts and bolts of new ventures, and the first task is to identify the motives that have brought the room together. Hadzima does not sort those motives into pure and impure kinds. He simply lays them out and lets us hear ourselves in them.

Some are here because they have never taken a formal class in entrepreneurship and want to see what the subject even looks like. Some are here because they have encountered some miserable product or process and want to make it less bad. Some have an invention and want to bring it to life. Some want to change the world. Some are drawn by the glamour and money that entrepreneurship still advertises.

The lecture's early tone matters. We are not yet being asked to judge a venture. We are being asked to notice why the possibility of a venture exerts a pull at all.

The Dropbox anecdote gives this opening its first concrete shape. A student, stuck without access to his files while traveling, does not merely complain. He converts irritation into a design imperative: make a storage system that does not "suck." The lecture uses this story carefully. It is not yet a theory of ventures. It is an example of how dissatisfaction, if pursued, can become the beginning of disciplined entrepreneurial thought.

1.2 The Entrepreneur's First Questions

From motive, the lecture widens into uncertainty. Hadzima says that he will give us guideposts and asks us to carry his questions from session to session. The effect is cumulative. We do not begin with a neat framework because the subject does not first arrive to us neatly.

1. How do we start?
2. What, exactly, do we do first?
3. What problem are we solving?
4. Does the proposed solution really answer that problem?
5. Who cares about the idea, specifically, and who is the customer?
6. How will the venture make money?
7. How long will it take to get to market, what will it cost, and what resources will be required?
8. What legal entity, if any, should be formed?
9. How can the idea be protected, if protection is the right objective?
10. Will cofounders be needed, and how will the relationship be structured?
11. How will negotiation enter, with employees, consultants, customers, advisors, and investors?
12. How do we discover what we do not know?

The lecture then adds two important qualifications. First, entrepreneurs are always negotiating because they are assembling people and resources they do not yet fully control in order to pursue a vision they do not yet fully own. Second, mistakes are not the scandal. Repeating the same mistake is the scandal. The people teaching the course, he says, have scars, and part of their use is that they may help us recognize recurring potholes before we drive directly into them.

By this point the chapter has already adopted the lecture's rhythm. We begin from personal motives, but we are not left in autobiography. We are pushed outward into operational questions. The subject becomes practical before it becomes formal.

1.3 Why Do This If Most Startups Fail?

At exactly the point where the lecture might have become merely energizing, Hadzima turns and creates resistance. Why do this at all? Why enter a line of work defined by difficulty, embarrassment, and attrition?

The success image is easy to summon: fame, fortune, public recognition. The lecture then opposes to that image the much harsher testimony of startup life. The Nvidia anecdote serves this purpose. The founder, looking back across decades, says in effect that if he had fully known what the road required, he might not have taken it. The point is not anti-entrepreneurial. It is corrective. We are meant to feel the cost before the course offers any technique for improving our odds.

The lecture gives the base rates in blunt numerical form:

$$\Pr(\text{startup success}) \approx 0.10 \approx 10\%, \tag{1.1}$$

$$\Pr(\text{startup failure}) \approx 0.90 \approx 90\%, \tag{1.2}$$

$$\Pr(\text{top-tier VC success per investment}) \approx \frac{3}{10} = 0.300. \tag{1.3}$$

The baseball analogy is part of the argument, not a decorative aside. A venture capitalist succeeding three times out of ten is likened to a .300 hitter: someone who fails often and is still a star. The analogy has a pedagogical use. It takes a discouraging ratio and translates it into a more familiar domain where high failure and excellence already coexist.

One slide, cited in the lecture, sharpens the point further by attributing most failure to poor planning and poor experience. We should not over-read the exact number attached to that slide, but the argumentative purpose is clear. Hadzima is preparing the answer to the question he has just raised. If failure were purely random, there would be little reason for a course like this. If planning and preparation matter, then a course may genuinely improve the odds.

That is why he then introduces the local MIT counterweight. The Kauffman-style numbers are not given as a proof that ventures succeed here routinely; they are given as evidence that when ventures launched in this environment do succeed, they may succeed at very large scale:

$$\frac{26{,}000}{120{,}000} \approx 0.217 \approx \frac{1}{5}. \tag{1.4}$$

On the lecture's telling, that means roughly one active MIT-founded company for every five living alumni in the period covered by the report. The same framing attributes annual revenues on the order of $2 trillion to those companies and says that, if aggregated, they would amount to roughly the eleventh largest economy in the world.

So the lecture does not deny the ugliness of the odds. It accepts them. Then it narrows the ambition. The course will not make venture formation safe. It will try to improve the probability of landing on the successful side of an intrinsically asymmetric game.

1.4 Why This Course, and Who Is In The Room?

After the odds have been stated baldly, the lecture repairs confidence by explaining what kind of course this is. It is not about theory. It is about doing. The speakers are not academics who study entrepreneurship from a distance; they are people who either are doing the work or have already done it.

The course's origin story reinforces the same point. Students effectively demanded a January course on how to start a company. Hadzima, short on time, assembled practitioners and asked them for the most useful possible format: tell the room the three or four or five things you wish somebody had told you when you started. That origin story is itself a small entrepreneurial parable. Demand appeared first; the product was assembled in response.

Then the lecture narrows from the course as a whole to the room immediately in front of it. This matters because the first equation in the lecture does not appear on a pure mathematics slide. It appears on a slide about audience heterogeneity: students and non-student participants, different backgrounds, different interest groups, different expectations.

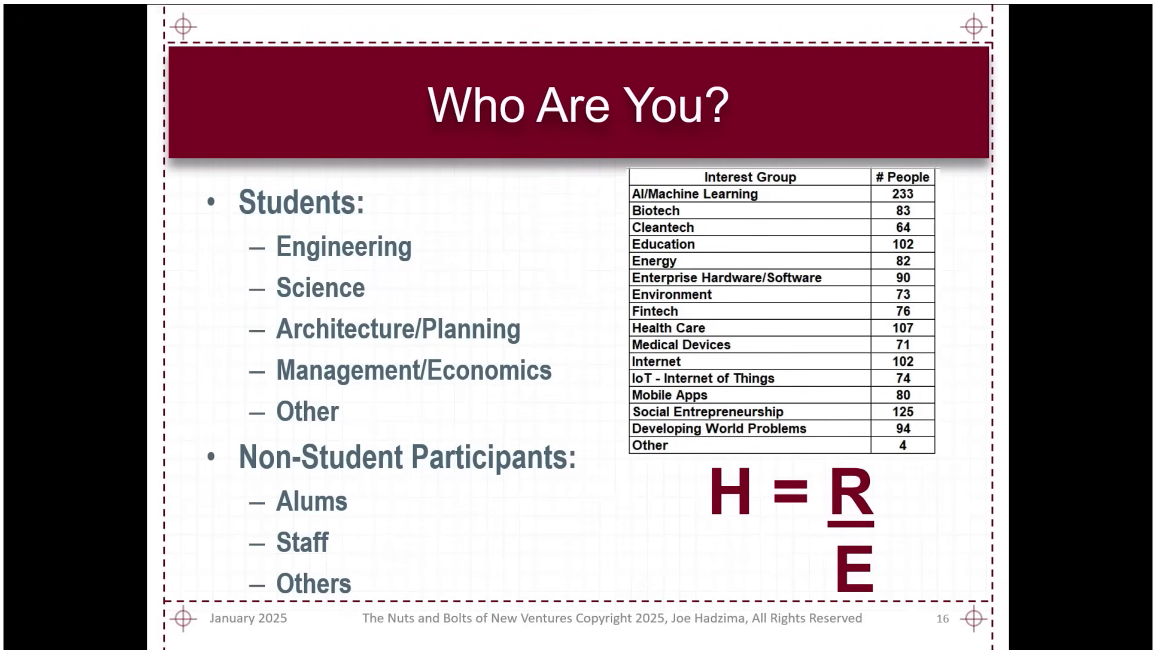

Figure 1.1: Audience mix and the course happiness equation.

The slide is really a room map. The audience breakdown at left and the small interest-group table at right explain why the equation belongs here. The lecture needs a compact rule for thinking about how one course can satisfy people who arrive with different prior knowledge and different hopes.

The slide writes the quotient in a stylized stacked form. We standardize it in ordinary mathematical notation:

$$H = \frac{R}{E}. \tag{1.5}$$

Definition 1.1 (The Happiness Equation)**.** In the lecture's notation,

$$H = \frac{R}{E},$$

where H denotes happiness, R denotes reality, and E denotes expectations.

Hadzima then states the course's objective in deliberately mathematical shorthand. The goal, he says, is to maximize H. We should hear that as lecture rhetoric rather than as a literal optimization problem, but the rhetoric is useful because it forces us to ask what in the ratio can be responsibly changed.

1.4.1 Question & Answer

Question. What are R and E in $H = \frac{R}{E}$?

Answer. R is reality and E is expectations. The lecture briefly records a revealing false start: some Sloan students suggest that R must be revenue and E must be expenses. Hadzima accepts that reading as a subset and then enlarges it. The equation is not an accounting identity. It is a more general rule for disappointment and satisfaction.

Once we write it cleanly, the lecture's verbal logic becomes immediate:

$$H = \frac{R}{E}, \tag{1.6}$$

$$H > 1 \iff R > E, \tag{1.7}$$

$$H = 1 \iff R = E, \tag{1.8}$$

$$H < 1 \iff R < E. \tag{1.9}$$

So if we underpromise and overdeliver, then $R > E$, and the ratio exceeds one. If we overpromise and underdeliver, then $R < E$, and the ratio falls below one. The lecture's little classroom equation is therefore already doing more than managing expectations for a single course. It is establishing a general rule about promises, delivery, and emotional aftermath.

Remark 1.2. The lecture jokes that if we could drive expectations to zero, we might produce infinite happiness. We should not literalize the joke into a usable model. The mathematical content we actually need is comparative: if expectations rise while reality does not, the quotient worsens.

This is also why the equation returns at the end of the lecture, after the discussion of pitch decks and Theranos. It is not a throwaway line. It is one of the lecture's organizing devices.

1.5 Frameworks for Judging a Venture

Now that the room has been calibrated, Hadzima begins compressing entrepreneurship into reusable filters. The first filter is almost embarrassingly simple, and that is part of its strength.

A venture must create value. If it creates no value, it has no subject matter. But creation alone is not enough. It must also capture enough of that value to continue. In the for-profit setting, that capture eventually takes the form of profit and cash flow. In the not-for-profit setting, it may take the form of enough attention, support, and incoming resources to keep operating. The structural distinction is important. Hadzima is careful not to reduce entrepreneurship to one legal or financial form.

The next compression is the "three whys," followed by a fourth question that appears only after the first three have begun to work:

1. **Why this?**
2. **Why now?**
3. **Why this team?**
4. **Why won't this work?**

Why this? The lecture's strongest illustration is Amy Smith's water-incubator pitch. Its power begins with scale: 1.9 billion people lacking access to clean water. Only after that scale has been felt does the mechanism appear. Biological contamination testing requires incubation. Existing incubators require electricity. The affected population frequently lacks electricity. Therefore a non-electric incubator attacks a real bottleneck. The structure is exact: large problem, narrow obstacle, specific bridge.

Why now? The FAA example gives this question its form. A technical solution may be genuinely good and still not be timely. But if institutional demand appears at the same moment that the solution becomes available, then timing and opportunity converge. The lecture's point is severe: five years early and five years late may both be failures even when the idea itself is sound.

Why this team? Prior experience can help, but it is not magic. Hadzima's Encore Computer example makes that clear. A team may have astonishing prior credentials and still fail. Prior experience can attract funding; it does not guarantee strategic coherence or executional success. For early founders without that prior experience, the question becomes sharper: what model, what judgment, what surrounding talent gives us reason to believe that this team is the right team?

Why won't this work? This question does not replace the first three. It arrives after them. Hadzima even frames it in sales language: by the time a prospect begins seriously probing limitations, interest already exists. That is why this fourth question belongs to de-risking. We are now asking not whether the venture sounds attractive, but whether its failure modes are understood well enough that we can plan around them.

By this point the lecture has moved from personal motive to evaluative machinery. But it has not yet become abstract. Each filter is anchored in a story, and each story exists to sharpen the filter rather than to replace it.

1.6 Ideas, Execution, Timing, and People

Hadzima says he dislikes frameworks, and then, sensibly, gives us one. The next compression is the lecture's central four-part model of venture success: ideas, execution, timing, and people. The lecture does something subtle here. It first presents the four-part structure in a general, almost static form, and only afterward begins to make one part of it dynamic.

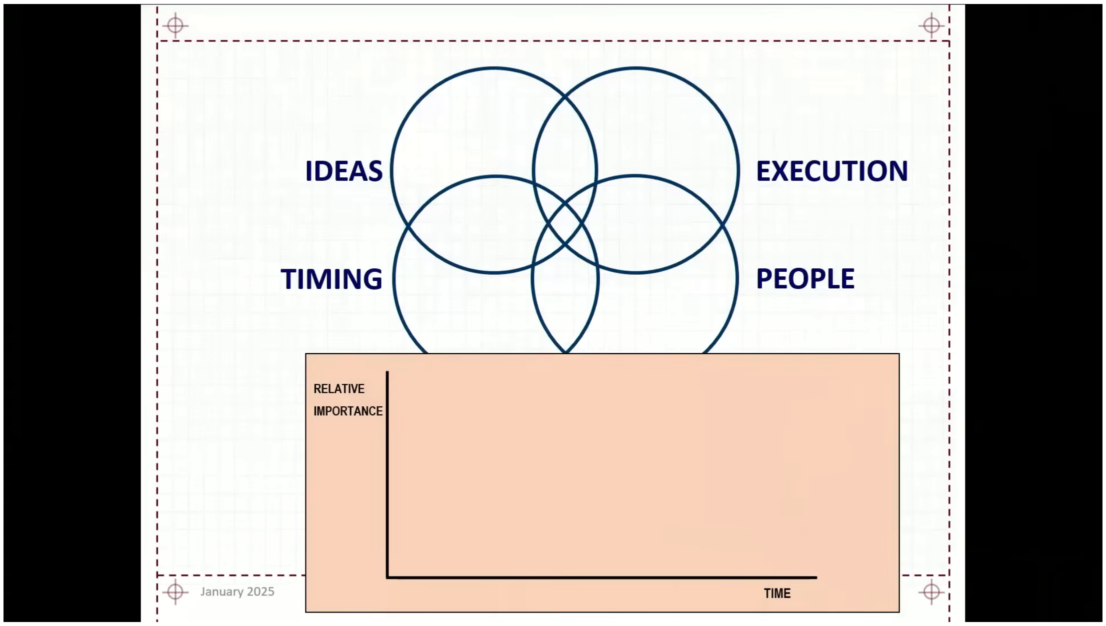

Figure 1.2: Four startup factors above an empty importance-time graph.

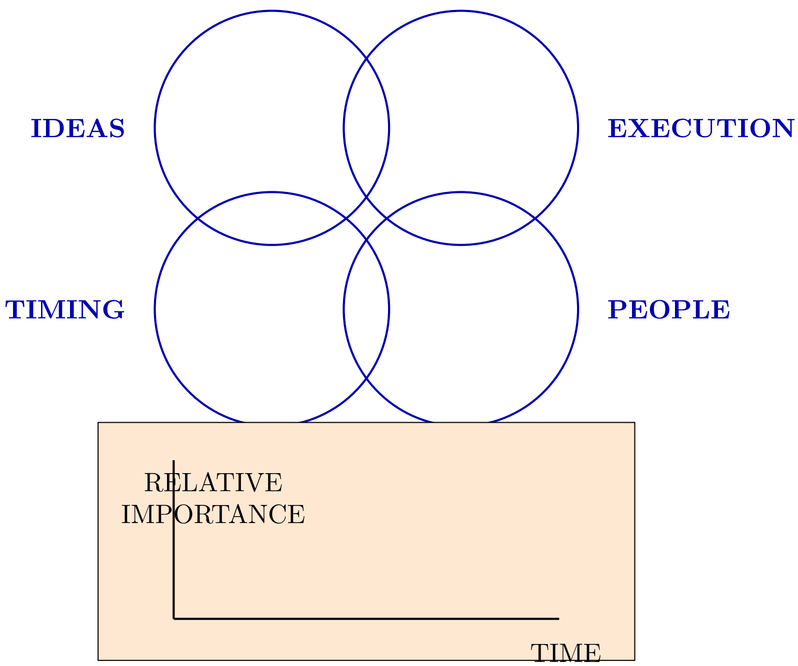

Figure 1.3: Pocket redraw of the four-factor framework before the lower graph is interpreted.

The upper panel says simultaneous alignment. The lower panel says, in effect, not yet. Although the lecture verbally introduces it as a "curve graph," the visible slide at this stage is only a scaffold: axes, labels, and empty space. That is exactly right narratively. The lower panel is being reserved for a specific problem that has not yet been solved.

1.6.1 Ideas

The lecture is impatient with the mere existence of ideas. Around MIT, ideas are everywhere. The real question is whether an idea is valuable, to whom it is valuable, how much value it creates, and whether that value can be defended long enough to support a venture. An idea that can be copied instantly may still be admirable; it may even be socially useful. But the lecture insists that admiration and venture viability are not the same thing.

1.6.2 Execution

Execution is where the lecture becomes particularly concrete. Hadzima uses examples of deals he passed on not to boast about discernment, but to show how execution uncertainty feels before an outcome is known.

Zipcar looked conceptually important. Renting a car by the hour is a genuine break in the consumption model of transportation. But the operational burden, especially around parking and city-by-city implementation, was for him obscure enough to justify passing. The precursor-to-eBay story works the same way. The market concept was visible early, but the trust and fraud machinery required for large-scale execution was not yet persuasive.

This is why the lecture pauses over *mens et manus*. Ideas and execution belong together. Yet even that pair is not sufficient.

1.6.3 Timing

Timing is treated with unusual seriousness. Hadzima says plainly that he has lost more money and time by being ahead of the curve than in many other ways. Timing, in this lecture, is not a matter of trendiness. It is a matter of whether the surrounding world has become compatible with the venture.

3D printing is the clean example. The technology was not "new" in the simple sense. What had changed was the supporting environment: materials, sensors, microelectronics, and the expiration of earlier constraints. Prodigy, by contrast, spent heavily to do online shopping before the user environment and browser infrastructure were ready. Fusion makes the point in an even harder form. It does not work in pieces. All of it must work together, and that makes timing inseparable from the completeness of the system.

So when the lecture asks "Why now?", it is not asking for a slogan about urgency. It is asking whether the world and the venture have arrived at the same time.

1.6.4 People

People, Hadzima says, are the single biggest source of failure in most ventures. He first frames this with the Cheshire Cat story: if we do not know where we are going, route choice is meaningless. He then makes the point vivid with the three-founder example. One founder wants the technology to become a standard, perhaps even open source. Another wants a fast-growing company that can go public. A third wants something important but more collegial and less intense. These are not three versions of the same ambition. They are three different journeys mistakenly housed in one venture.

The lecture then adds two more people problems. Sometimes teams are funded before they understand what they are doing, and then the pivot becomes internal rather than customer-facing. And sometimes teams are too certain too early. The E-Ink example belongs here. Not knowing what we do not know is itself a people problem because it determines whether a team is teachable.

1.6.5 Question & Answer

Question. How do technical and business contributions change over time inside a founding team?

Answer. The lecture answers this with a qualitative time-dependent graph. Early on, the technical side has very high relative importance because until something works technically, there is nothing to finance, market, or scale. The business founder, by contrast, may initially be working nights and weekends, perhaps still holding another job, trying to understand the market and the eventual commercial structure. Over time the balance changes. The business side rises in importance because distribution, team building, financing, negotiation, and organization become increasingly central.

The lecture's clean symbolic summary is qualitative rather than quantitative:

$$I_{\text{technical}}(t)\downarrow, \qquad I_{\text{business}}(t)\uparrow\,. \tag{1.10}$$

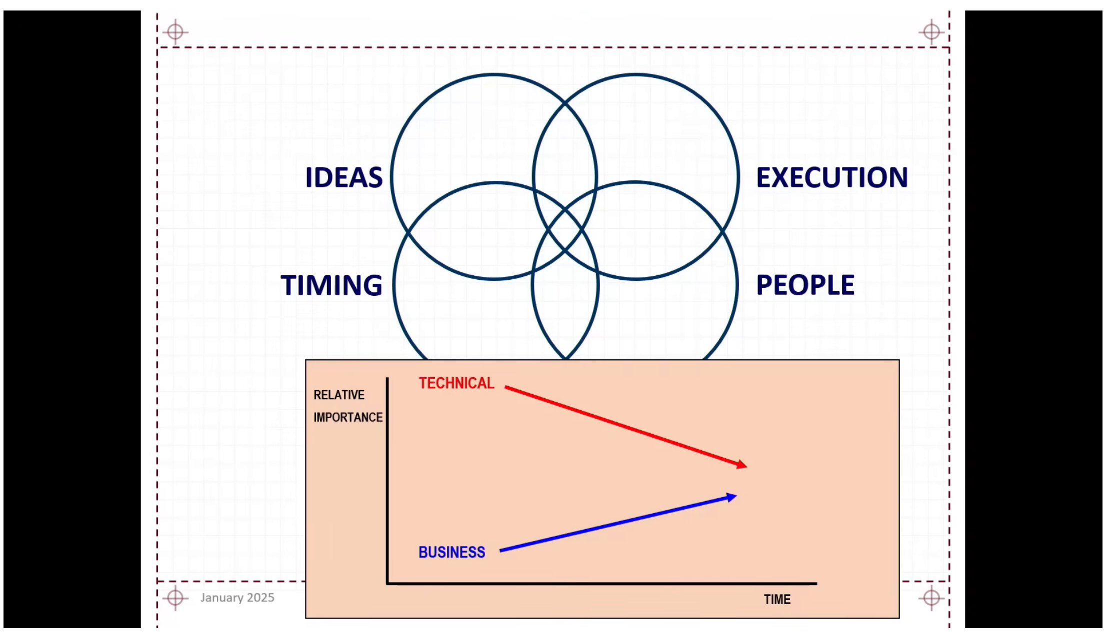

Figure 1.4: Technical importance falls as business importance rises.

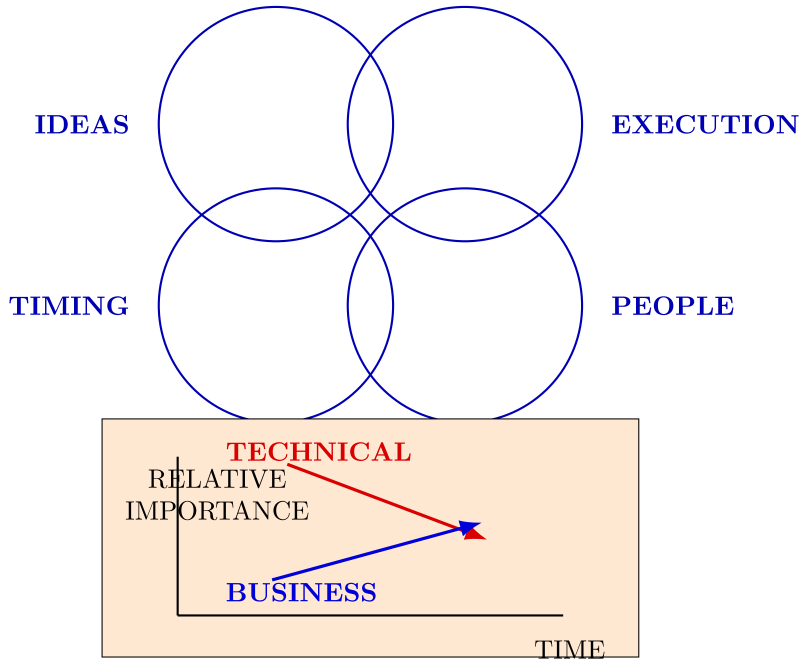

Figure 1.5: Pocket redraw of the lecture's qualitative team-balance graph.

Remark 1.3. The graph is schematic. The arrows are straight, unscaled, and not tied to a marked crossing time. The lecture explicitly says that whether the lines cross is not the point. The point is that the relative balance changes, and with it the conversation about legitimacy, structure, and equity.

Once the graph has opened the team problem, the lecture does not leave the point in diagrammatic form. It immediately turns to actual ventures and asks which of the four components aligned and which failed.

SpeechCo. This is the lecture's success case of alignment. The idea was strong. The timing was good because compute power had finally arrived. Execution improved by restricting the system to a domain of language use rather than trying to solve everything at once. And the people had spent enough time together before launch to form a functioning team.

VideoCo. This case initially looks like an execution failure in the making. The founders seem incapable even of choosing a company name. The payoff is that they were not merely naming the company; they were choosing a strategy. The DIVA/Avid connection reveals that what looked like hesitation had a directional logic.

HIV Co. Here the idea, timing, and scientific credibility were all strong. The weakness was people. Cofounder conflict drained the venture even in the presence of first-rate backers and advisers. This is Hadzima's cleanest case of a venture weakened primarily by internal human structure.

NanoCo. This is the lecture's timing case. NanoFuel had technical promise and environmental appeal, but the market arrived in the wrong price regime. The lecture even gives a back-of-theenvelope threshold:

$$p_{\text{diesel}} \approx \$0.50/\text{gal}, \tag{1.11}$$
$$p_{\text{neutral}} \approx \$2.00/\text{gal}. \tag{1.12}$$

At launch,

$$p_{\text{diesel}} < p_{\text{neutral}}, \tag{1.13}$$

so the product fell below the lecturer's rough economic-neutral point. The lecture makes the reasoning even more vivid by observing that one could not buy a gallon of distilled water for \$0.50. The argument is intentionally simple, but it is enough. A technically interesting product may fail because the world supplies the wrong price at the wrong time.

1.7 From Vision to Pitch, Without Losing the Substance

Only after all this does the lecture turn to communication. The order is essential. We are not being taught to substitute rhetoric for venture substance. We are being taught how substance is compressed when it must be explained to other people.

Hadzima begins with the mission statement or vision, sometimes also treated as a value proposition. He then immediately insists that the vision must be supported. The lecture's pyramid is a hierarchy of compression.

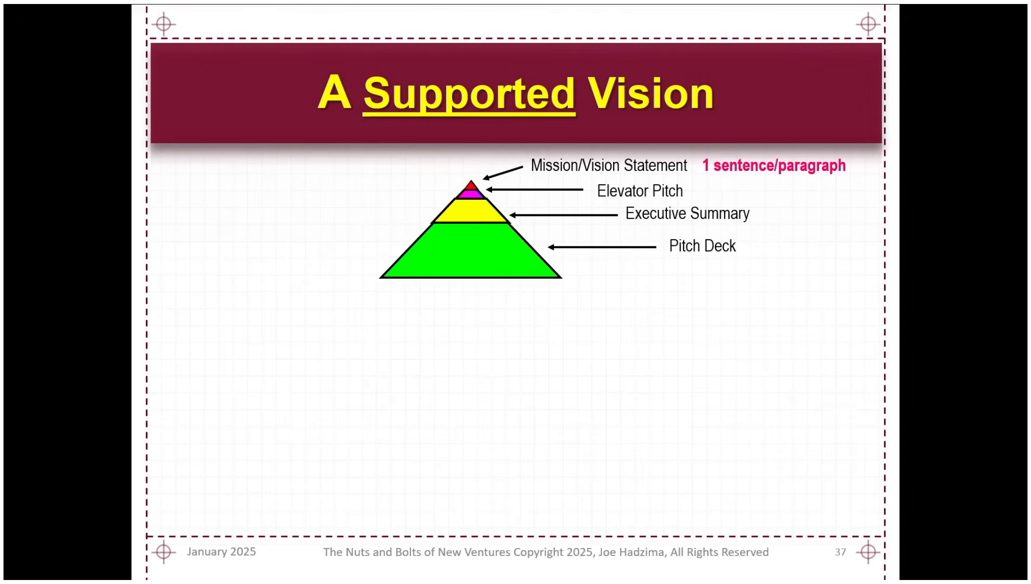

Figure 1.6: A supported vision from mission statement to pitch deck.

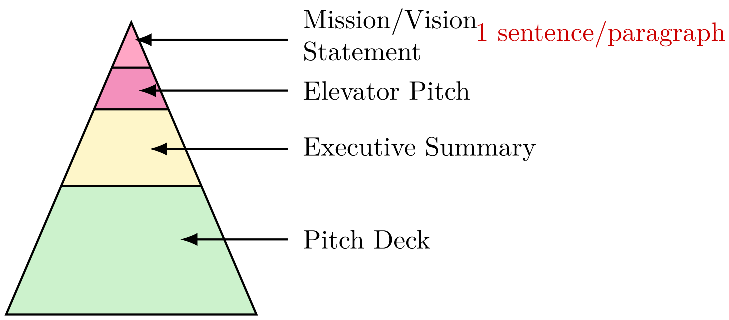

Figure 1.7: Pocket redraw of the supported-vision pyramid.

The original slide is worth keeping because the arrow targets fix the order of the tiers. The apex is the most compressed claim, and the lower tiers are progressively larger supports. One can read the entire device as repeated compression of the same venture at different bandwidths:

$$\text{mission statement} \sim 1 \text{ sentence or paragraph}, \tag{1.14}$$

$$\text{elevator pitch} \sim 30 \text{ seconds}, \tag{1.15}$$

$$\text{executive summary} \sim 1\text{–}4 \text{ pages}, \tag{1.16}$$

$$(\text{slides}, \text{minutes}, \text{font size}) = (10, 20, 30). \tag{1.17}$$

That last line is Guy Kawasaki's 10/20/30 rule: ten slides, twenty minutes, thirty-point font. Hadzima presents it as discipline. Each level asks the same question at a different scale: can the venture still make sense when compressed?

This is why the lecture then moves directly to the Theranos warning. Communication matters, but unsupported communication is not persuasion. It is deferred collapse. The lecture's earlier happiness equation becomes useful again. If reality is fixed while expectations are inflated, the ratio worsens:

$$R \text{ fixed}, \qquad E_2 > E_1 \qquad \Longrightarrow \qquad \frac{R}{E_2} < \frac{R}{E_1}. \tag{1.18}$$

So the lecture's final position is not anti-pitch. It is anti-separation. Sizzle without substance is empty, but substance that cannot be articulated will often fail to recruit the people, capital, and trust required for the venture to exist.

1.8 Summary

We can now see the lecture unfolding in the order Hadzima intended. It begins with motives, expands into operational ignorance, then creates the central tension by insisting on bad odds and painful realities. Only after that tension is fully present does it justify the course as a practical instrument for preparation. The room itself then becomes the site of the first equation, $H = \frac{R}{E}$, and that local rule about reality and expectations becomes a general rule about promises, delivery, and hype.

From there the lecture moves by successive compressions. First: create value and capture value. Then: why this, why now, why this team, and why won't this work? Then: ideas, execution, timing, and people. Finally: mission statement, elevator pitch, executive summary, and pitch deck. Each compression is followed by examples, because the lecture never allows a framework to float free of lived ventures.

The ending is modest, and that modesty is part of the seriousness of the lecture. Luck remains. Preparation does not abolish it. It changes the odds of being ready when luck matters. That is why the lecture ends not with startup triumphalism but with a practical handoff: break the room into interest groups, meet collaborators, return for logistics, and then go find customers.